

UNIT-3: MASS TRANSPORT

Total Mass Flux = Molecular Mass (or molar) Flux
(or molar) + Convective Mass (or molar) Flux

① Molecular Mass/Molar Flux Vector ($\underline{J}_A$ or $\underline{j}_A$)

Governed by Fick's Law of Diffusion

in molar units

$$\begin{aligned}\underline{J}_{A3} &= -D_{AB} \frac{dc_A}{dz} \left[\frac{\text{mol}}{\text{m}^2 \cdot \text{s}} \right] \\ &= -c D_{AB} \frac{dx_A}{dz}\end{aligned}$$

For binary system

$$n_A = \text{mol A}$$

$$n_B = \text{mol B}$$

$$\text{Total moles, } n = n_A + n_B$$

$$x_A = \frac{n_A}{n} \quad x_B = \frac{n_B}{n}$$

$$x_A + x_B = 1$$

$$c_A = \frac{n_A}{V} \quad c_B = \frac{n_B}{V}$$

$$c = c_A + c_B = \frac{n_A + n_B}{V} = \frac{n}{V}$$

$$c_A = \frac{n_A}{V} \cdot \frac{n}{n} = \frac{n_A}{n} \cdot \frac{n}{V}$$

$$\Rightarrow \boxed{c_A = x_A c}$$

in mass units

$$\begin{aligned}\underline{j}_{A3} &= -D_{AB} \frac{d\rho_A}{dz} \left[\frac{\text{kg}}{\text{m}^2 \cdot \text{s}} \right] \\ &= -\rho D_{AB} \frac{dw_A}{dz}\end{aligned}$$

For binary system

$$m_A = \text{mass A}$$

$$m_B = \text{mass B}$$

$$\text{Total mass, } m = m_A + m_B$$

$$w_A = \frac{m_A}{m} \quad w_B = \frac{m_B}{m}$$

$$w_A + w_B = 1$$

$$\rho_A = \frac{m_A}{V} \quad \rho_B = \frac{m_B}{V}$$

$$\rho = \rho_A + \rho_B = \frac{m_A + m_B}{V} = \frac{m}{V}$$

$$\rho_A = \frac{m_A}{V} \cdot \frac{m}{m} = \frac{m_A}{m} \cdot \frac{m}{V}$$

$$\Rightarrow \boxed{\rho_A = w_A \rho}$$

② Convective Mass Flux Vector ($\rho_A \underline{V}$)

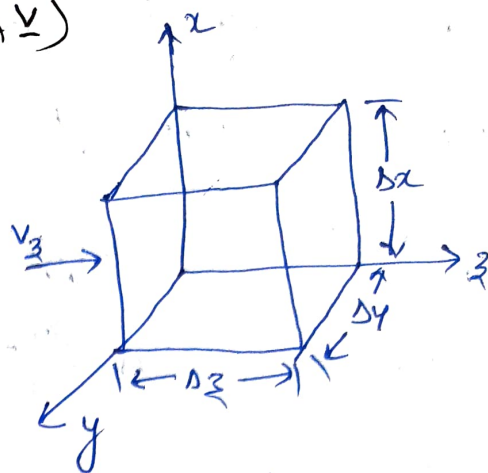
Volumetric flowrate across $\Delta x \Delta y$
 $= v_z (\Delta x \Delta y)$

Mass flowrate across $\Delta x \Delta y$
 of A $= \rho_A v_z (\Delta x \Delta y)$

Mass flux $= \rho_A v_z$ (in z-direction)
 of A

$\therefore$ convective mass flux vector $= \rho_A \underline{V}$

similarly, convective molar flux vector $= c_A \underline{V}$



Total Molar Flux

$$\underline{N}_A = \underline{J}_A + c_A \underline{V} \quad \left[\frac{\text{mol A}}{\text{m}^2 \cdot \text{s}} \right]$$

z-component:

$$N_{Az} = J_{Az} + c_A v_z$$

$$\Rightarrow N_{Az} = -c D_{AB} \frac{dx_A}{dz} + c_A v_z$$

Total Mass Flux

$$\underline{n}_A = \underline{j}_A + \rho_A \underline{V} \quad \left[\frac{\text{kg A}}{\text{m}^2 \cdot \text{s}} \right]$$

z-component:

$$n_{Az} = j_{Az} + \rho_A v_z$$

$$\Rightarrow n_{Az} = -\rho D_{AB} \frac{dw_A}{dz} + \rho_A v_z$$

For Binary System of A & B

$$N_{Az} = -c D_{AB} \frac{dx_A}{dz} + c_A v_z \quad \text{①}$$

$$N_{Bz} = -c D_{AB} \frac{dx_B}{dz} + c_B v_z \quad \text{②}$$

Adding ① & ②,

$$N_{Az} + N_{Bz} = -c D_{AB} \frac{d}{dz} (x_A + x_B) + (c_A + c_B) v_z$$

$$\Rightarrow N_{Az} + N_{Bz} = -c D_{AB} \frac{d}{dz} (1) + c v_z$$

$$\Rightarrow N_{Az} + N_{Bz} = \frac{c_A}{x_A} v_z \quad \left[\because c_A = x_A c \right]$$

$$\Rightarrow c_A v_z = x_A (N_{Az} + N_{Bz})$$

Substituting in eq. ①

$$N_{A_z} = -c D_{AB} \frac{dx_A}{dz} + x_A (N_{A_z} + N_{B_z}) \quad (3)$$

If B is non-diffusing, $N_{B_z} = 0$

$$\Rightarrow N_{A_z} = -c D_{AB} \frac{dx_A}{dz} + x_A N_{A_z}$$

$$\Rightarrow (1 - x_A) N_{A_z} = -c D_{AB} \frac{dx_A}{dz}$$

$$\Rightarrow N_{A_z} = - \frac{c D_{AB}}{1 - x_A} \frac{dx_A}{dz} \quad (4)$$

Steady state Mass Balance

$$\left[\begin{array}{c} \text{Rate of} \\ \text{mass of A} \\ \text{IN} \end{array} \right] - \left[\begin{array}{c} \text{Rate of mass} \\ \text{of A OUT} \end{array} \right] + \left[\begin{array}{c} \text{Rate of production} \\ \text{of mass of A by} \\ \text{Homogeneous chemi-} \\ \text{-cal Reaction} \end{array} \right] = 0$$

Boundary conditions

① Concentration at a surface may be specified, eg. $x_A = x_{A0}$

② Mass/molar flux at a surface, eg. $N_{A_z} = N_{A0}$

③ If ~~diffusion~~ a solid is diffusing into surrounding fluid (gas/liquid) stream, then at fluid-solid interface

$$N_{A0} = k_c (C_{A0} - C_{Ab})$$

C_{A0} = surface concentration of A

C_{Ab} = concentration of A in bulk fluid.

k_c = mass transfer coefficient

④ Rate of chemical reaction at a surface.

① Diffusion through a stagnant Gas Film

Liquid "A" is diffusing into gas "B".

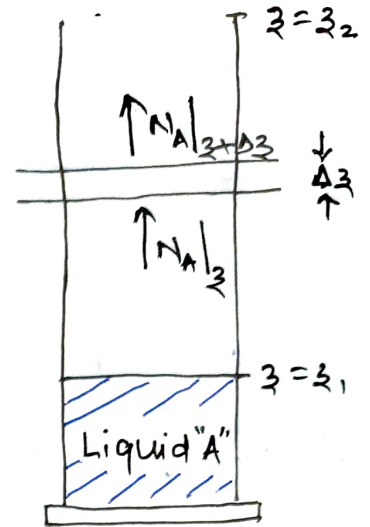
Gas (A+B)

Assumptions:

- ① Both A & B are ideal gases.
- ② A device is used to maintain constant liquid level $z = z_1$
- ③ B is non-diffusing
- ④ At steady state

$$\text{At } z = z_1: \quad x_A = x_{A1}$$

$$\text{At } z = z_2: \quad x_A = x_{A2}$$



Consider a shell of thickness Δz

Cross-sectional area of tube = S

Steady state mass balance of A over shell:

$$S N_{A3}|_z - S N_{A3}|_{z+\Delta z} = 0.$$

$$\Rightarrow \frac{1}{\Delta z} \left[N_{A3}|_{z+\Delta z} - N_{A3}|_z \right] = 0$$

$$\Rightarrow \frac{d}{dz} N_{A3} = 0$$

$$\Rightarrow \frac{d}{dz} \left(- \frac{c D_{AB}}{1 - x_A} \frac{dx_A}{dz} \right) = 0$$

{ Using Eq. ①

$$\Rightarrow \int d \left(\frac{1}{1 - x_A} \frac{dx_A}{dz} \right) = \int 0 \cdot dz$$

$$\Rightarrow \frac{1}{1 - x_A} \frac{dx_A}{dz} = C_1$$

Integrate again

$$\int -\frac{1}{1-x_A} \frac{d(1-x_A)}{dz} = \int C_1$$

$$\Rightarrow -\ln(1-x_A) = C_1 z + C_2$$

$$\text{Let } C_1 = -\ln k_1 \quad \& \quad C_2 = -\ln k_2$$

$$\Rightarrow -\ln(1-x_A) = -(z \ln k_1 + \ln k_2)$$

$$\Rightarrow \ln(1-x_A) = \ln k_1^z + \ln k_2 = \ln \frac{k_1^z}{k_2} \ln(k_1^z k_2)$$

$$\Rightarrow 1-x_A = k_1^z k_2 \quad \text{(A1)}$$

$$\text{B.C. (1): At } z=z_1: \quad x_A = x_{A1} \Rightarrow 1-x_{A1} = k_1^{z_1} k_2 \quad \text{(A2)}$$

$$\text{B.C. (2): At } z=z_2: \quad x_A = x_{A2} \Rightarrow 1-x_{A2} = k_1^{z_2} k_2 \quad \text{(A3)}$$

$$\frac{\text{Eq. (A3)}}{\text{Eq. (A2)}} \Rightarrow \frac{1-x_{A2}}{1-x_{A1}} = \frac{k_1^{z_2} k_2}{k_1^{z_1} k_2} = k_1^{(z_2-z_1)}$$

$$\Rightarrow k_1 = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{1}{z_2-z_1}}$$

$$\frac{\text{Eq. (A1)}}{\text{Eq. (A2)}} \Rightarrow \frac{1-x_A}{1-x_{A1}} = \frac{k_1^z k_2}{k_1^{z_1} k_2} = k_1^{(z-z_1)}$$

$$\Rightarrow \boxed{\frac{1-x_A}{1-x_{A1}} = \left(\frac{1-x_{A2}}{1-x_{A1}} \right)^{\frac{z-z_1}{z_2-z_1}}}$$

$$\Rightarrow \frac{x_B}{x_{B1}} = \left(\frac{x_{B2}}{x_{B1}} \right)^\gamma \quad \text{(A4)}$$

$$\text{where } \gamma = \frac{z-z_1}{z_2-z_1} \Rightarrow (z_2-z_1)\gamma = z-z_1$$

$$\Rightarrow \frac{d\gamma}{dz} = \frac{1}{z_2-z_1}$$

$$\text{for } z=z_1 \Rightarrow \gamma=0$$

$$\text{for } z=z_2 \Rightarrow \gamma=1$$

① Average concentration

$$\left(\frac{x_B}{x_{B1}}\right)_{avg} = \frac{\int_{z_1}^{z_2} a^\gamma dz}{\int_{z_1}^{z_2} dz}$$

Let $\frac{x_{B2}}{x_{B1}} = a$

$$\Rightarrow (x_B)_{avg} = \frac{x_{B1}}{z_2 - z_1} \int_{z_1}^{z_2} a^\gamma dz = \frac{x_{B1}}{z_2 - z_1} \int_0^1 a^\gamma \left(\frac{dz}{d\gamma}\right) d\gamma$$

$$= x_{B1} \int_0^1 a^\gamma d\gamma$$

$$= x_{B1} \left[\frac{a^\gamma}{\ln a} \right]_0^1$$

$$= \frac{x_{B1}}{\ln a} (a - 1)$$

$$= \frac{x_{B1}}{\ln \left(\frac{x_{B2}}{x_{B1}}\right)} \left(\frac{x_{B2}}{x_{B1}} - 1 \right)$$

$$\left\{ \because \int a^x dx = \frac{a^x}{\ln a} \right.$$

$$\Rightarrow \boxed{(x_B)_{avg} = \frac{x_{B2} - x_{B1}}{\ln \left(\frac{x_{B2}}{x_{B1}}\right)}}$$

(A5)

② Rate of Mass Transfer at Gas-Liquid interface, $N_{A3}|_{z_1}$

From eq. (4)

$$N_{A3}|_{z_1} = - \frac{C D_{AB}}{1 - x_{A1}} \frac{dx_A}{dz} \Big|_{z_1}$$

(A6)

Let us first find out $\frac{dx_B}{dz} \Big|_{z_1}$

differentiating eq. (A4)

$$\frac{dx_B}{dz} = x_{B1} \frac{d}{dz} \left(\frac{x_{B2}}{x_{B1}} \right)^\gamma = x_{B1} \frac{d}{d\gamma} \left(\frac{x_{B2}}{x_{B1}} \right)^\gamma \frac{d\gamma}{dz}$$

$$= \frac{x_{B1}}{z_2 - z_1} \frac{d}{d\gamma} a^\gamma = \frac{x_{B1}}{z_2 - z_1} a^\gamma \ln a$$

$$= \frac{x_{B1}}{z_2 - z_1} \left(\frac{x_{B2}}{x_{B1}} \right)^\gamma \ln \left(\frac{x_{B2}}{x_{B1}} \right)$$

$$\Rightarrow \left. \frac{dx_B}{dz} \right|_{z=z_1} = \frac{x_{B1}}{z_2 - z_1} \ln \left(\frac{x_{B2}}{x_{B1}} \right) \quad \left\{ \because \text{for } z=z_1, \gamma=0 \right.$$

Eq (A6) can be written as

$$N_{A2} \Big|_{z=z_1} = \frac{c D_{AB}}{x_{B1}} \left. \frac{dx_B}{dz} \right|_{z=z_1} = \frac{c D_{AB}}{x_{B1}} \frac{x_{B1}}{z_2 - z_1} \ln \left(\frac{x_{B2}}{x_{B1}} \right)$$

$$\Rightarrow \boxed{N_{A2} \Big|_{z=z_1} = \frac{c D_{AB}}{z_2 - z_1} \ln \left(\frac{x_{B2}}{x_{B1}} \right)} \quad \text{(A7)}$$

Problems based on (A) Diffusion through a stagnant gas film.

- ① Calculate the mass flux of Benzene through a layer of air 10mm thickness at 25°C & 200 kN/m^2 (total pressure), partial pressure of Benzene is 6 kN/m^2 at the left side of the layer and 1 kN/m^2 at the right side. $D_{AB} = 4.4 \times 10^{-6}\text{ m}^2/\text{s}$.

Soln: Given:

$$z_2 - z_1 = 10\text{ mm} = 0.01\text{ m}$$

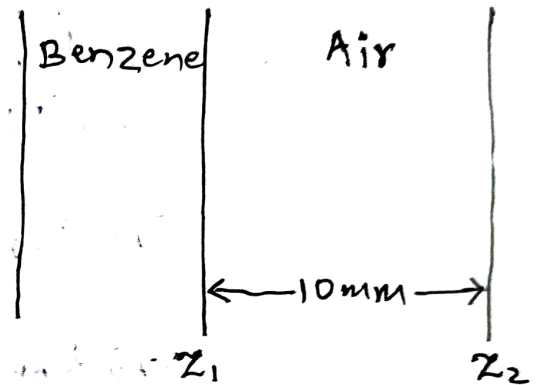
$$T = 25^\circ\text{C} = 298\text{ K}$$

$$P = 200\text{ kPa}$$

$$P_{A1} = 6\text{ kPa}$$

$$P_{A2} = 1\text{ kPa}$$

$$D_{AB} = 4.4 \times 10^{-6}\text{ m}^2/\text{s}$$



Let $A \rightarrow$ Benzene & $B \rightarrow$ Air

To Find: $N_{AZ} \Big|_{z=z_1}$

Assumption: Air-Benzene vapor mixture follows Ideal Gas Law.

$$\therefore PV = nRT$$

$$\Rightarrow c = \frac{n}{V} = \frac{P}{RT}$$

Using Eq. (A7)

$$N_{AZ} \Big|_{z_1} = \frac{c D_{AB}}{z_2 - z_1} \ln \left(\frac{x_{B2}}{x_{B1}} \right) = \frac{P D_{AB}}{RT (z_2 - z_1)} \ln \left(\frac{x_{B2}}{x_{B1}} \right)$$

$$\text{Also, } \frac{x_{B2}}{x_{B1}} = \frac{P_{B2}/P}{P_{B1}/P} = \frac{P_{B2}}{P_{B1}}$$

$$\therefore N_{AZ} \Big|_{z_1} = \frac{P D_{AB}}{RT(z_2 - z_1)} \ln \left(\frac{P_{B2}}{P_{B1}} \right)$$

$$P_{B2} = P - P_{A2} = 199 \text{ kPa.}$$

$$P_{B1} = P - P_{A1} = 194 \text{ kPa}$$

$$\begin{aligned} \Rightarrow N_{AZ} \Big|_{z_1} &= \frac{2 \times 10^5 \text{ Pa} \times 4.4 \times 10^{-6} \text{ m}^2/\text{s}}{8314 \text{ Pa} \cdot \text{m}^3/(\text{kgmol} \cdot \text{K}) \times 298 \text{ K} \times 0.01 \text{ m}} \ln \left(\frac{199}{194} \right) \\ &= (9.04 \times 10^{-7}) \text{ kgmol}/(\text{m}^2 \cdot \text{s}) \end{aligned}$$

② What is the rate of evaporation of chloropicrin into air at 298 K? Total pressure = 770 mm Hg, diffusivity = $0.088 \text{ cm}^2/\text{s}$, vapor pressure = 23.81 mm Hg, $z_2 - z_1 = 11.14 \text{ cm}$, density of chloropicrin = 1.65 g/cm^3 , $S = 2.29 \text{ cm}^2$.

Soln: Given:

$$T = 298 \text{ K}$$

$$P = 770 \text{ mm Hg}$$

$$D_{AB} = 0.088 \text{ cm}^2/\text{s} = 8.8 \times 10^{-6} \text{ m}^2/\text{s}$$

$$S = 2.29 \text{ cm}^2 = 2.29 \times 10^{-4} \text{ m}^2$$

$$z_2 - z_1 = 11.14 \text{ cm} = 0.1114 \text{ m}$$

$$P_{A1} = 23.81 \text{ mm Hg}$$

$$P_{B1} = P - P_{A1} = 746.19 \text{ mm Hg}$$

$$P_{A2} = 0 \Rightarrow P_{B2} = 770 \text{ mm Hg}$$

Using Eq. (A7)

$$\begin{aligned} N_{A2}|_{z_1} &= \frac{c D_{AB}}{z_2 - z_1} \ln \left(\frac{x_{B2}}{x_{B1}} \right) = \frac{P D_{AB}}{RT (z_2 - z_1)} \ln \left(\frac{p_{B2}}{p_{B1}} \right) \\ &= \frac{770 \text{ mmHg} \times 8.8 \times 10^{-6} \text{ m}^2/\text{s}}{62.36 \text{ mmHg} \cdot \text{m}^3/(\text{kgmol} \cdot \text{K}) \times 298 \text{ K} \times 0.1114 \text{ m}} \ln \left(\frac{770}{746.19} \right) \\ &= (1.028 \times 10^{-7}) \text{ kgmol}/(\text{m}^2 \cdot \text{s}) \end{aligned}$$

$$\text{Rate of Evaporation} = S N_{A2}|_2$$

$$= 2.29 \times 10^{-4} \text{ m}^2 \times 1.028 \times 10^{-7} \frac{\text{kgmol}}{\text{m}^2 \cdot \text{s}}$$

$$= 2.354 \times 10^{-11} \frac{\text{kgmol}}{\text{s}}$$

⑧ Diffusion with a Heterogeneous Chemical Reaction
 Assume a Dimerization reaction $2A \rightarrow B$ is taking place on a catalyst surface

$$N_{Bz} = -\frac{1}{2} N_{Az}$$

substituting in eq. (3)

$$N_{Az} = -c D_{AB} \frac{dx_A}{dz} + \frac{1}{2} x_A N_{Az}$$

$$\Rightarrow N_{Az} = - \frac{c D_{AB}}{1 - \frac{x_A}{2}} \frac{dx_A}{dz}$$

(B1)

Edge of Hypothetical

Assumptions:

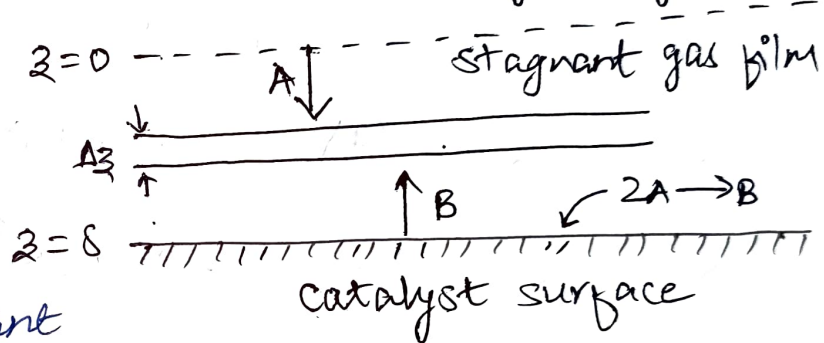
① Reaction $2A \rightarrow B$ is instantaneous & irreversible.

② A hypothetical stagnant gas film exists between catalyst surface & turbulent gas stream (A+B).

③ At steady state

$$\text{At } z=0: \quad x_A = x_{A0} \quad \text{or} \quad x_B = x_{B0}$$

$$\text{At } z=\delta: \quad x_A = 0$$



consider a thin shell of thickness Δz within Hypothetical gas film

steady state Mass Balance of A over shell:

$$S N_{Az} \Big|_z - S N_{Az} \Big|_{z+\Delta z} = 0$$

$$\Rightarrow \frac{1}{\Delta z} \left[N_{Az} \Big|_{z+\Delta z} - N_{Az} \Big|_z \right] = 0$$

Taking Limit $\Delta z \rightarrow 0$

$$\Rightarrow \frac{d}{dz} N_{A2} = 0 \Rightarrow \frac{d}{dz} \left(- \frac{c D_{AB}}{1 - \frac{x_A}{2}} \frac{dx_A}{dz} \right)$$

$$\Rightarrow \frac{d}{dz} \left(\frac{1}{1 - \frac{x_A}{2}} \frac{dx_A}{dz} \right) = 0$$

Integrating w.r.t. z

$$\Rightarrow \frac{1}{1 - \frac{x_A}{2}} \frac{dx_A}{dz} = C_1$$

Integrating again w.r.t. z

$$\Rightarrow -2 \ln \left(1 - \frac{x_A}{2} \right) = C_1 z + C_2$$

$$\text{Let } C_1 = -2 \ln K_1 \quad \& \quad C_2 = -2 \ln K_2$$

$$\Rightarrow -2 \ln \left(1 - \frac{x_A}{2} \right) = (-2 \ln K_1) z - 2 \ln K_2$$

$$\Rightarrow -2 \ln \left(1 - \frac{x_A}{2} \right) = -2 \ln (K_1^z K_2)$$

$$\Rightarrow 1 - \frac{x_A}{2} = K_1^z K_2 \quad \textcircled{B2}$$

$$\text{B.C. (1): At } z=0, \quad x_A = x_{A0} \Rightarrow 1 - \frac{x_{A0}}{2} = K_2 \quad \textcircled{B3}$$

$$\text{B.C. (2): At } z=\delta, \quad x_A = 0 \Rightarrow 1 = K_1^\delta K_2$$

$$\Rightarrow K_1 = \left(\frac{1}{K_2} \right)^{\frac{1}{\delta}} = K_2^{-\frac{1}{\delta}}$$

$$= \left(1 - \frac{x_{A0}}{2} \right)^{-\frac{1}{\delta}} \quad \textcircled{B4}$$

Substituting in $\textcircled{B2}$

$$1 - \frac{x_A}{2} = \left(1 - \frac{x_{A0}}{2} \right)^{-\frac{z}{\delta}} \left(1 - \frac{x_{A0}}{2} \right)$$

$$\Rightarrow \boxed{1 - \frac{x_A}{2} = \left(1 - \frac{x_{A0}}{2} \right)^{1 - \frac{z}{\delta}}} \quad \textcircled{B5}$$

Differentiating w.r.t. z

$$-\frac{1}{2} \frac{dx_A}{dz} = \left(1 - \frac{x_{A0}}{2}\right)^{-\frac{2}{\delta}}$$

$$-\frac{1}{2} \frac{dx_A}{dz} = \left(1 - \frac{x_{A0}}{2}\right)^{1 - \frac{2}{\delta}} \cdot \ln\left(1 - \frac{x_{A0}}{2}\right) \cdot \left(-\frac{1}{\delta}\right)$$

$$\Rightarrow \frac{dx_A}{dz} = \frac{2}{\delta} \left(1 - \frac{x_{A0}}{2}\right)^{1 - \frac{2}{\delta}} \ln\left(1 - \frac{x_{A0}}{2}\right)$$

$$\Rightarrow \left. \frac{dx_A}{dz} \right|_{z=0} = \frac{2}{\delta} \left(1 - \frac{x_{A0}}{2}\right) \ln\left(1 - \frac{x_{A0}}{2}\right)$$

Molar Flux of A at the Edge of Hypothetical stagnant gas film:

$$N_{A2} \Big|_{z=0} = - \frac{c D_{AB}}{1 - \frac{x_{A0}}{2}} \left. \frac{dx_A}{dz} \right|_{z=0}$$

$$= -c D_{AB} \frac{2}{\delta} \ln\left(1 - \frac{x_{A0}}{2}\right)$$

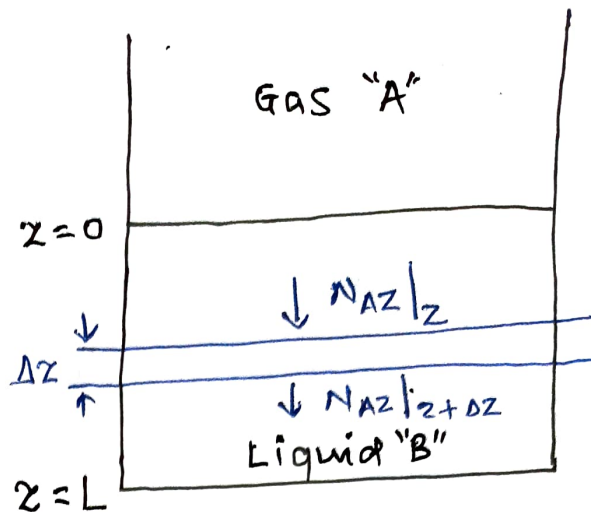
$$= \frac{2 c D_{AB}}{\delta} \ln\left(\frac{2}{2 - x_{A0}}\right)$$

(B6)

③ DIFFUSION WITH A HOMOGENEOUS CHEMICAL REACTION

- Gas A dissolves in Liquid B & dissolves isothermally into the liquid phase.

- "A" undergoes an irreversible first order homogeneous chemical reaction



Rate of reaction is given by $r_A = -k C_A$ $\left[\frac{\text{mol}}{\text{m}^3 \cdot \text{s}} \right]$

consider a small shell of thickness Δz

Assumptions:

① Mol fraction of A in liquid phase is very low & can be neglected, i.e. $x_A \approx 0$

$$\Rightarrow N_{AZ} = -C D_{AB} \frac{dx_A}{dz} = -D_{AB} \frac{dC_A}{dz} \quad \text{①}$$

② At steady state, $x_A = x_{A0}$ $C_A = C_{A0}$ at $z=0$.

Steady state Mass (or mol) balance over shell:

$$\left[\text{Rate of mol "A" IN} \right] - \left[\text{Rate of mol "A" OUT} \right] - \left[\text{Rate of consumption of "A" in chemical reaction} \right] = 0$$

$$\Rightarrow S N_{AZ}|_z - S N_{AZ}|_{z+\Delta z} - (S \Delta z) k C_A = 0$$

$\Rightarrow$ Divide by $S \Delta z$ & rearrange

$$\frac{1}{\Delta z} \left[N_{AZ}|_{z+\Delta z} - N_{AZ}|_z \right] + k C_A = 0$$

Taking limit $\Delta z \rightarrow 0$

$$\Rightarrow \frac{dN_A z}{dz} + k c_A = 0$$

$$\Rightarrow -D_{AB} \frac{d^2 c_A}{dz^2} + k c_A = 0$$

$$\Rightarrow \frac{d^2 c_A}{dz^2} + -\frac{k}{D_{AB}} c_A = 0$$

$$\Rightarrow \frac{d^2 c_A}{dz^2} - E c_A = 0$$

(C2)

$$\left\{ \text{Let } E = \frac{k}{D_{AB}} \right.$$

$$\text{Let } c_A = e^{rz}$$

$$\Rightarrow \frac{dc_A}{dz} = r e^{rz}$$

(First derivative)

$$\Rightarrow \frac{d^2 c_A}{dz^2} = r^2 e^{rz}$$

(second derivative)

Substituting in (C2)

$$r^2 e^{rz} - E e^{rz} = 0$$

$$\Rightarrow r^2 = E \Rightarrow r = \pm \sqrt{E} = \pm \sqrt{\frac{k}{D_{AB}}}$$

$$\therefore c_A = c_1 e^{\sqrt{E}z} + c_2 e^{-\sqrt{E}z}$$

$$= c_1 e^{Fz} + c_2 e^{-Fz}$$

(C3)

$$\left\{ \text{Let } F = \sqrt{E} \right.$$

B.C.①: At $z=0$, $c_A = c_{A0}$

substituting in (C3)

$$c_{A0} = c_1 + c_2 \Rightarrow c_2 = c_{A0} - c_1$$

B.C.②: At $z=L$, $N_{A2} = 0 \Rightarrow \left. \frac{dc_A}{dz} \right|_{z=L} = 0$

$$\Rightarrow (c_1 F e^{Fz} - c_2 F e^{-Fz}) \Big|_{z=L} = 0$$

$$\Rightarrow c_1 e^{FL} - c_2 e^{-FL} = 0$$

$$\Rightarrow C_1 e^{FL} - (C_{A0} - C_1) e^{-FL} = 0$$

$$\Rightarrow C_1 e^{FL} - C_{A0} e^{-FL} + C_1 e^{-FL} = 0$$

$$\Rightarrow C_1 = C_{A0} \frac{e^{-FL}}{e^{FL} + e^{-FL}}$$

$$\Rightarrow C_2 = C_{A0} - C_1 = C_{A0} \left[1 - \frac{e^{-FL}}{e^{FL} + e^{-FL}} \right] = C_{A0} \frac{e^{FL}}{e^{FL} + e^{-FL}}$$

Substituting C_1 & C_2 in Eq. (C3)

$$\begin{aligned} \frac{C_A}{C_{A0}} &= \frac{e^{-FL} e^{Fz}}{e^{FL} + e^{-FL}} + \frac{e^{FL} e^{-Fz}}{e^{FL} + e^{-FL}} = \frac{e^{F(z-L)} + e^{-F(z-L)}}{e^{FL} + e^{-FL}} \\ &= \frac{\cosh[F(z-L)]}{\cosh(FL)} = \frac{\cosh\left[\sqrt{\frac{k}{D_{AB}}}(z-L)\right]}{\cosh\left(L\sqrt{\frac{k}{D_{AB}}}\right)} \end{aligned}$$

(i) Average concentration

$$\begin{aligned} [C_A]_{\text{avg}} &= \frac{\int_0^L C_A dz}{\int_0^L dz} = \frac{1}{L} \int_0^L C_A dz \\ &= \frac{C_{A0}}{L \cosh(FL)} \int_0^L \cosh[F(z-L)] dz \\ &= \frac{C_{A0}}{FL \cosh(FL)} \left[\sinh[F(z-L)] \right]_0^L \\ &= \frac{C_{A0}}{FL \cosh(FL)} [\sinh(0) - \sinh(-FL)] \\ &= \frac{C_{A0} \sinh(FL)}{FL \cosh(FL)} = \frac{C_{A0}}{FL} \tanh(FL) \\ &= \frac{C_{A0}}{L \left(\frac{k}{D_{AB}}\right)^{1/2}} \tanh\left(L\sqrt{\frac{k}{D_{AB}}}\right) \end{aligned}$$

(ii) Molar Flux at GL interface

$$N_{AZ} \Big|_{z=0} = -D_{AB} \frac{dc_A}{dz} \Big|_{z=0}$$

$$= - \frac{C_{A0} D_{AB}}{\cosh(FL)} \frac{d}{dz} \{ \cosh[F(z-L)] \} \Big|_{z=0}$$

$$= - \frac{C_{A0} D_{AB} F}{\cosh(FL)} \left[\sinh F(z-L) \right]_{z=0}$$

$$= \frac{C_{A0} D_{AB} F}{\cosh(FL)} \sinh(FL)$$

$$= C_{A0} D_{AB} F \tanh(FL)$$

$$= C_{A0} D_{AB} \sqrt{\frac{k}{D_{AB}}} \tanh \left(L \sqrt{\frac{k}{D_{AB}}} \right)$$

$$= C_{A0} \sqrt{k D_{AB}} \tanh \left(L \sqrt{\frac{L}{D_{AB}}} \right)$$

Diffusion Through Pinex Tube

Helium diffuses through pinex tube containing natural gas

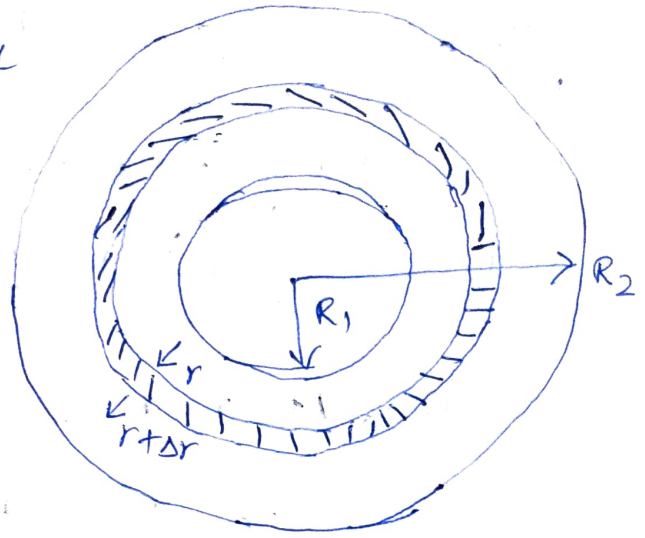


R_1 = Radius of inner tube

R_2 = Outer radius of tube

$$\text{At } r = R_1: C_A = C_{A1}$$

$$\text{At } r = R_2: C_A = C_{A2}$$



eq. (1): $N_{Ar} = -D_{AB} \frac{dC_A}{dr}$ (only diffusion, no convection)

Consider a thin cylindrical shell of thickness Δr .

$$\text{Rate of A in across surface at } r = 2\pi L (r N_{Ar}) \Big|_r$$

$$\begin{aligned} \text{Rate of A out across surface at } r + \Delta r &= 2\pi L (r + \Delta r) N_{Ar} \Big|_{r+\Delta r} \\ &= 2\pi L (r N_{Ar}) \Big|_{r+\Delta r} \end{aligned}$$

Steady state mass balance of A over thin section

$$2\pi L (r N_{Ar}) \Big|_r - 2\pi L (r N_{Ar}) \Big|_{r+\Delta r} = 0$$

$$\Rightarrow \lim_{\Delta r \rightarrow 0} \frac{d}{dr} (r N_{Ar}) = 0$$

$$\Rightarrow \frac{d}{dr} (r N_{Ar}) = 0 \Rightarrow \int d(r N_{Ar}) = \int 0 \cdot dr$$

$$\Rightarrow r N_{Ar} = C_1 \Rightarrow N_{Ar} = \frac{C_1}{r} \Rightarrow -D_{AB} \frac{dC_A}{dr} = \frac{C_1}{r}$$

$$\Rightarrow \int dC_A = -\frac{C_1}{D_{AB}} \int \frac{dr}{r}$$

$$\Rightarrow C_A = -\frac{C_1}{D_{AB}} \ln r + C_2$$

(2)

B.C. ①: At $r=R_1$, $C_A = C_{A1}$

Substituting in eq. ②,

$$C_{A1} = - \frac{C_1}{D_{AB}} \ln R_1 + C_2 \quad (3)$$

B.C. ②: At $r=R_2$, $C_A = C_{A2}$

Substituting in eq. ②

$$C_{A2} = - \frac{C_1}{D_{AB}} \ln R_2 + C_2 \quad (4)$$

$$(3) - (4) \Rightarrow C_{A1} - C_{A2} = \frac{C_1}{D_{AB}} (\ln R_2 - \ln R_1) = \frac{C_1}{D_{AB}} \ln \left(\frac{R_2}{R_1} \right)$$

$$\Rightarrow \frac{C_1}{D_{AB}} = \frac{C_{A1} - C_{A2}}{\ln \left(\frac{R_2}{R_1} \right)}$$

$$(2) - (4) \Rightarrow C_A - C_{A2} = \frac{C_1}{D_{AB}} (\ln R_2 - \ln r) = \frac{C_1}{D_{AB}} \ln \left(\frac{R_2}{r} \right)$$

$$\Rightarrow C_A = C_{A2} + \frac{C_{A1} - C_{A2}}{\ln \left(\frac{R_2}{R_1} \right)} \ln \left(\frac{R_2}{r} \right)$$

Differentiating w.r.t. r

$$\frac{dC_A}{dr} = \frac{C_{A1} - C_{A2}}{\ln \left(\frac{R_2}{R_1} \right)} \left(-\frac{1}{r} \right) = - \frac{C_{A1} - C_{A2}}{r \ln \left(\frac{R_2}{R_1} \right)}$$

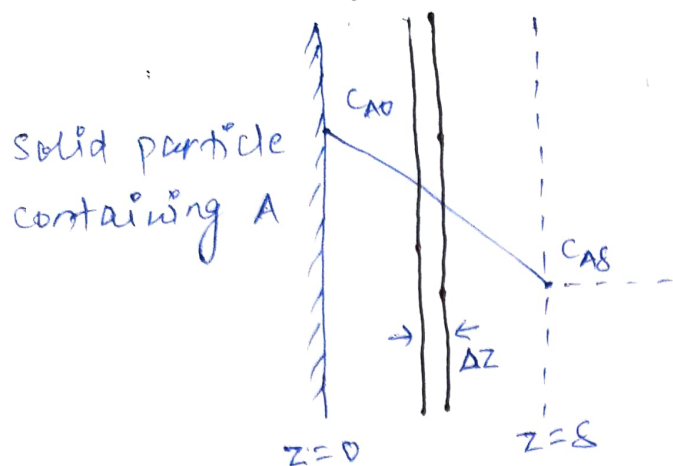
$$\Rightarrow N_{Ar} \Big|_{r=R_2} = D_{AB} \frac{C_{A1} - C_{A2}}{R_2 \ln \left(\frac{R_2}{R_1} \right)} \quad (\text{from eq. ①})$$

Rate of leakage of He at outer surface ($r=R_2$)

$$W_A = 2\pi L R_2 N_{Ar} \Big|_{r=R_2}$$

$$\Rightarrow \boxed{W_A = 2\pi L D_{AB} \frac{C_{A1} - C_{A2}}{\ln \left(\frac{R_2}{R_1} \right)}}$$

Rate of Leaching



A liquid solvent extracts component "A" from solid particles. We are interested in the region close to a particle.

Assumptions:

- ① curvature of particle is ignored.
 - ② Only diffusion, no convection
 - ③ There exists a stagnant liquid film in the vicinity of solid particle through which A diffuses.
- δ = Thickness of stagnant liquid film.

B.C. ①: At $z=0$, $C_A = C_{A0}$

B.C. ②: At $z=\delta$, $C_A = C_{A\delta}$

$$N_{Az} = -D_{AB} \frac{dC_A}{dz} \quad \text{①}$$

consider a thin slab of thickness Δz within the stagnant liquid film.

Rate of A IN across surface at $z = \delta N_{Az}|_z$

Rate of A OUT across surface at $z+\Delta z = \delta N_{Az}|_{z+\Delta z}$

steady state mass balance of A over thin slab

$$\delta N_{Az}|_z - \delta N_{Az}|_{z+\Delta z} = 0$$

$$\Rightarrow \lim_{\Delta z \rightarrow 0} \frac{1}{\Delta z} \left[N_{Az} \Big|_{z+\Delta z} - N_{Az} \Big|_z \right] = 0$$

$$\Rightarrow \frac{dN_{Az}}{dz} = 0 \Rightarrow dN_{Az} = 0 \cdot dz$$

Integrating once

$$\int dN_{Az} = \int 0 \cdot dz \Rightarrow N_{Az} = C_1$$

$$\Rightarrow -D_{AB} \frac{dc_A}{dz} = C_1 \Rightarrow dc_A = -\frac{C_1}{D_{AB}} dz$$

Integrating once again

$$\int dc_A = -\frac{C_1}{D_{AB}} \int dz \Rightarrow C_A = -\frac{C_1}{D_{AB}} z + C_2 \quad (2)$$

substituting B.C. (1) in eq. (2)

$$C_{A0} = -\frac{C_1}{D_{AB}} (0) + C_2 \Rightarrow C_2 = C_{A0}$$

substituting B.C. (2) in eq. (2)

$$C_{AS} = -\frac{C_1}{D_{AB}} s + C_{A0} \Rightarrow -\frac{C_1}{D_{AB}} = \frac{C_{AS} - C_{A0}}{s}$$

substituting in eq. (2)

$$C_A = \frac{C_{AS} - C_{A0}}{s} z + C_{A0}$$

$$\Rightarrow \frac{dc_A}{dz} = \frac{C_{AS} - C_{A0}}{s}$$

Molar flux of Leaching

$$N_{Az} = -D_{AB} \frac{dc_A}{dz} = D_{AB} (C_{A0} - C_{AS}) / s$$

Q. Explain the procedure for setting up & solving mass flow problems.

Ans. ① After analyzing the flow/geometry/configuration carefully, write the expression for Total Molar Flux (N_A) or Total mass flux (m_A).

Step ② consider a differential volume (or shell) within flow or domain where mass transfer is taking place.

Step ③ Identify different surfaces of the shell (or small section) across which mass transfer is taking place.

Step ④ Apply Law of Conservation of Mass over the differential volume:

④ Unsteady state (For non-reacting medium):

$$[\text{Rate of mass (or moles) in}] - [\text{Rate of mass (or moles) out}] = [\text{Rate of change of mass (or moles)}]$$

⑥ Unsteady state For homogeneous chemical Reaction medium

$$\begin{aligned} & \left[\begin{array}{c} \text{Rate of} \\ \text{mass/mole} \\ \text{IN} \end{array} \right] - \left[\begin{array}{c} \text{Rate of} \\ \text{mass/mole} \\ \text{OUT} \end{array} \right] + \left[\begin{array}{c} \text{Rate of} \\ \text{Generation} \end{array} \right] - \left[\begin{array}{c} \text{Rate of} \\ \text{consumption} \end{array} \right] \\ & = \left[\begin{array}{c} \text{Mass/Molar} \\ \text{Rate of change} \end{array} \right] \end{aligned}$$

For Steady State Analysis, substitute this in

④ & ⑥: $\left[\begin{array}{c} \text{mass/molar} \\ \text{Rate of change} \end{array} \right] = 0$

step

⑤ Formulate steady/Unsteady state mass balance using

$$\left[\begin{array}{l} \text{Rate of mass (or mole) IN} \\ \text{or OUT across a surface} \end{array} \right] = \left[\begin{array}{l} \text{Area of} \\ \text{surface} \end{array} \right] \times \left[\begin{array}{l} N_A \text{ or } n_A \\ \text{across that} \\ \text{surface} \end{array} \right]$$

step ⑥ Boundary conditions

After formulating the steady/unsteady state mass balance equation for the differential volume, this equation is then integrated in order to derive concentration profile (x_A or C_A).

To remove integration constants, boundary conditions are to be identified & used.

Possible boundary conditions are

- ① concentration at a surface, ~~$N_A = N_{Ao}$~~ $x_A = x_{Ao}$
- ② Mass/Molar flux at a surface, $N_{Az} = N_{Ao}$
- ③ Rate of chemical reaction at a surface
- ④ If diffusion is occurring in a solid medium, then at solid-fluid interface,

$$N_{Ao} = k_c (C_{Ao} - C_{Ab})$$

where,

N_{Ao} = molar flux across interface

C_{Ao} = concentration at interface

C_{Ab} = concentration in bulk fluid.

k_c = Mass transfer coefficient.

Once the integration constants are removed, concentration profile can be achieved in terms of x_A or C_A .